

Supplementary Material

How Predictable Are Rare Precipitation Extremes? A Calibration-Aware Assessment for London and Paris

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S1 Results at the 99th-percentile threshold

The main paper analyses the 95th-percentile index, which has 417 extreme days (287 train, 130 test). The 99th-percentile index is far scarcer — only 137 extreme days in thirty years (101 train, 36 test) — which precludes the held-out intensity analysis (that requires extreme days in both train and test across several splits). For completeness we report here the results of the full model set at the 99th percentile, including the GP–EVT hurdle. The pattern of the main paper repeats: the GP–EVT hurdle is again the best model, while the more flexible response distributions (notably the zero-inflated log-normal) fall well below climatology.

Table S1 shows the bake-off (seed-averaged over three seeds, horizon $h = 1$), scored against climatology as in the main paper. The qualitative picture matches the 95th percentile but is weaker and noisier, as expected with a third as many events:

- **Occurrence remains weakly predictable.** A plain calibrated logistic regression with the ERA5 drivers beats climatology (Brier skill score $+0.015$), improving on index history alone ($+0.001$); the gain again comes chiefly from the convective-instability and circulation fields. Resolution (informativeness) is positive but smaller than at the 95th percentile (0.36×10^{-3} versus 0.83×10^{-3}).
- **The distributional models fall below climatology.** With only 101 training exceedances, the zero-inflated gamma and eGPD regressions are slightly worse than climatology on the Brier score, and the zero-inflated log-normal is again catastrophically miscalibrated — echoing, more sharply, the main-paper finding that the response distribution matters and that data scarcity penalises the more flexible models.
- **The GP–EVT hurdle wins again.** The GP–EVT hurdle is the best model at the 99th percentile too — the highest Brier skill ($+0.027$, above logistic’s $+0.015$) and the *only* model with positive threshold-weighted CRPS skill ($+0.016$) — echoing its main-paper win, though the margin is slim at this data volume.
- **The tail is all but climatology-bound.** Apart from the GP hurdle’s slim positive margin, no model beats the climatological tail on the threshold-weighted CRPS (all other skill scores ≤ 0), and the scarce exceedances do not support the held-out intensity test.

In short, the 99th-percentile threshold tells the same story as the 95th — weak but genuine occurrence predictability led by convective instability, a tail only the GP–EVT hurdle edges past, and a penalty on over-flexible response distributions (the zero-inflated log-normal) — but with too few events to support the intensity analysis or to add to the main conclusions.

Table S1: Bake-off at the 99th-percentile threshold (London, $h = 1$, seed-averaged). Brier skill score (BSS) and threshold-weighted CRPS skill (twCRPSS) are relative to climatology; positive is better. RES is the Brier resolution ($\times 10^3$). Climatology and a stationary peaks-over-threshold model define the reference. The GP-EVT hurdle is the same model as in the main paper (Section 4.2); best per column in bold. Compare with Table 2 of the main paper (95th percentile).

Model	BSS	RES $\times 10^3$	twCRPSS
Climatology / stationary POT	0.000	0.00	0.000
Logistic (index history)	+0.001	0.08	—
Logistic (+ERA5)	+0.015	0.36	—
ZI gamma (+ERA5)	−0.069	0.20	−0.130
ZI extended-GPD (+ERA5)	−0.048	0.26	−0.087
ZI log-normal (+ERA5)	−1.153	0.25	−4.827
GP-EVT hurdle (+ERA5)	+0.027	0.19	+0.016

S2 Paris: per-city diagnostic figures

The main paper shows the data, calibration, and horizon-decay diagnostics (Figures 1, 2, and 4 there) for London as the representative city, while the core comparative results — the bake-off, the held-out intensity analysis, and the unified hurdle — are reported for both cities. For completeness we give the Paris counterparts of the three single-city diagnostic figures here. They are qualitatively identical to London: a zero-inflated, right-skewed, short-memory index with a summer convective peak (Figure S1); well-calibrated exceedance probabilities from the simple logistic model and the GP-EVT hurdle, with the zero-inflated log-normal systematically overconfident (Figure S2); and occurrence skill decaying toward the prevalence floor with lead time while conformal coverage stays at the nominal level (Figure S3). Paris differs from London only in having a higher extreme-day prevalence (a more convective regime), consistent with its stronger predictability in the main text.

S3 Feature importance and horizon-wise occurrence skill

The main paper reports, in prose, that antecedent convective-instability and circulation indices dominate the occurrence feature screen (Section 5.3), and that occurrence skill decays with forecast horizon (Section 5.6, Figure 4). We give both results here as tables, to make them directly reproducible rather than only visually apparent.

Table S2 shows the 15 most frequently selected features (of $d = 49$; Section 3.3) under L_1 -logistic stability selection (200 subsamples of 60% of the training data, selection frequency = share of subsamples in which a feature’s coefficient is non-zero). The K-index of convective instability and the most recent index lag (`lag_1`) are selected in essentially every subsample in both cities, consistent with the main text; day-of-year seasonality, integrated moisture/vapour-transport fields, the east–west mean-sea-level-pressure gradient, and the North Atlantic Oscillation (NAO) index also recur near the top of both lists.

Table S3 gives the Brier skill score (relative to climatology) of the logistic +ERA5 occurrence model at each horizon of the sweep $h \in \{1, 3, 7, 14, 28\}$ (single fit per horizon; RES is the Brier resolution, $\times 10^3$). This is the proper-score counterpart of the discrimination-only (PR-AUC) horizon figure in the main paper and Supplementary Figure S3: in London, skill is effectively confined to $h = 1$ and collapses toward zero by $h = 3$; in Paris — consistent with its larger, more convective sample — skill remains small but positive out to $h = 28$, though it decays steadily from its $h = 1$ value.

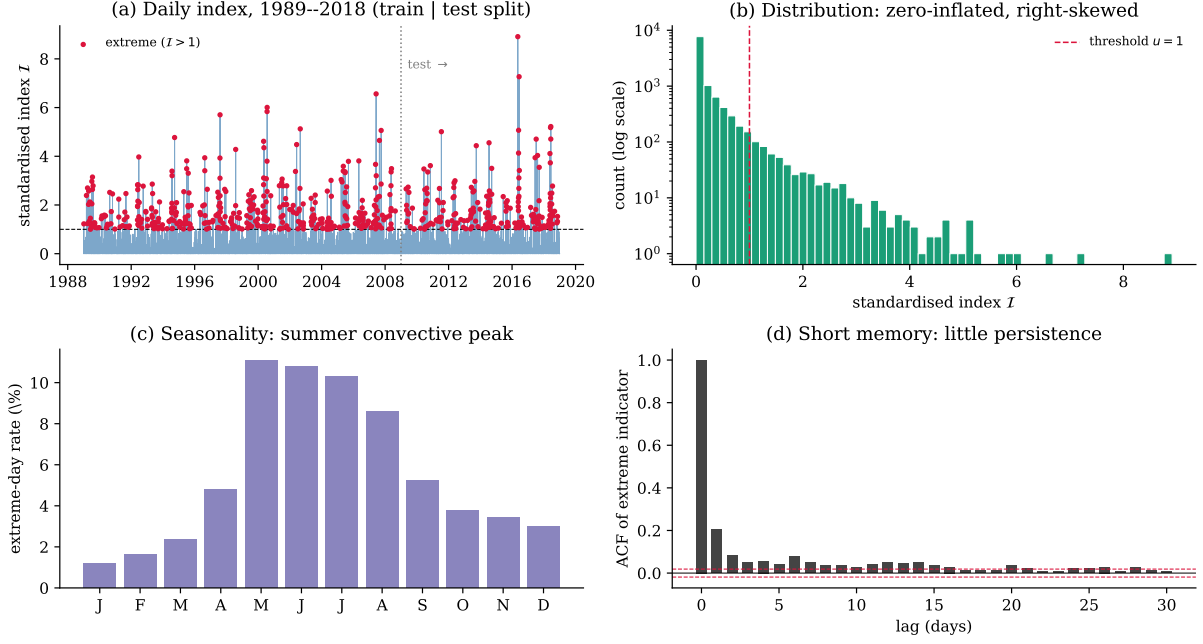


Figure S1: Paris 95th-percentile precipitation-extreme index (Paris counterpart of Figure 1 in the main paper). **(a)** Daily index with extreme days ($\mathcal{I} > 1$) and the train|test split; **(b)** zero-inflated, right-skewed distribution; **(c)** summer-peaked seasonality of the extreme-day rate; **(d)** short memory of the extreme indicator.

S4 The dependent-hurdle coupling: a null result

The GP-EVT hurdle of the main paper (Section 4.2) couples its occurrence and intensity latents through a linear model of coregionalisation, whose off-diagonal correlation ρ is a *borrowing-strength* parameter ($\rho = 0$ recovers the independent hurdle). The motivation is that, because the excess is observed only on the few extreme days, letting the abundant occurrence data inform the tail could stabilise intensity estimation. We test that mechanism directly here and find that, for this hurdle at this data volume, *coupling is weakly identified and neither helps nor hurts*; we therefore retain it only as a harmless nested generalisation of the independent hurdle, not as a source of skill.

Coupling does not change forecast skill. Re-fitting the 95th-percentile bake-off model (+ERA5, $h = 1$, seed-averaged over three seeds, 600 epochs) with the coupling switched off is indistinguishable from the coupled fit on both proper scores (Table S4): the tail-CRPS and Brier skills are identical in London and marginally *higher* without coupling in Paris. Whatever the GP-EVT hurdle achieves (Table 2 of the main paper), it owes nothing to the coregionalisation.

Recovery is weak on synthetic data. On data simulated from the generative model with a known occurrence–intensity correlation ($\xi = 0.20$; three seeds each), the coupled hurdle recovers ρ only weakly and noisily — $\hat{\rho} = 0.33 \pm 0.48$, 0.56 ± 0.25 , and 0.79 ± 0.05 at $\rho_{\text{true}} = 0.3$, 0.7 , and 0.9 respectively — and the tail-log-likelihood gain from coupling is negligible at every setting. Crucially, unlike an over-parameterised base model whose independent intensity head can collapse (so that coupling appears to rescue it), the single-layer GP hurdle used here is already stable: there is no instability for borrowing strength to cure, so the benefit coregionalisation is designed to provide does not materialise.

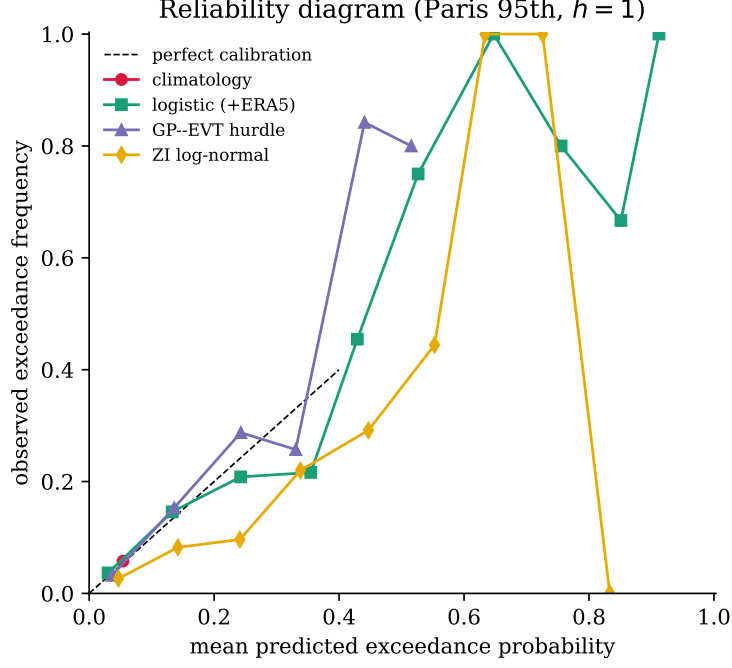


Figure S2: Reliability diagram for the Paris exceedance probability (95th, $h = 1$; Paris counterpart of Figure 2). The calibrated logistic model and the GP-EVT hurdle track the diagonal; the zero-inflated log-normal is systematically overconfident despite attaining nominal conformal coverage, mirroring London.

Real-data estimates. Fitting the coupled hurdle on the real series (five seeds) gives $\hat{\rho} = +0.55 \pm 0.16$ in London and $+0.38 \pm 0.04$ in Paris. Given the weak identification just documented, these cannot be read as measurements of real occurrence–intensity dependence; we report them for completeness and draw no conclusion about the presence or absence of exploitable coupling.

In short, the dependent-hurdle coupling is a sound nested generalisation of the independent hurdle but is empirically inert at this data volume: it is retained because it does no harm and reduces to the independent model as $\rho \rightarrow 0$, not because it improves forecasts.

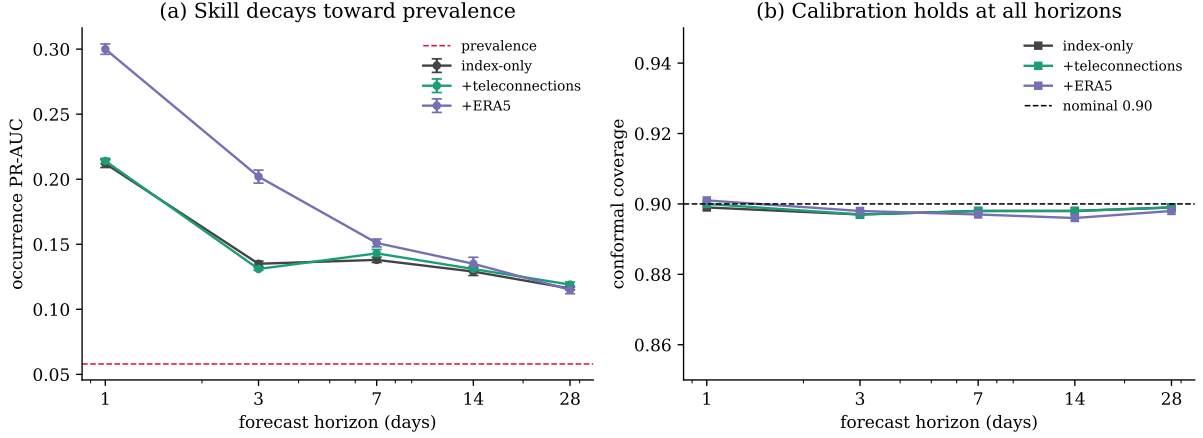


Figure S3: Seed-averaged horizon sweep on the Paris 95th-percentile index (Paris counterpart of Figure 4; GP-EVT hurdle). **(a)** Occurrence PR-AUC decays toward the prevalence floor (dashed) as the horizon grows, with the +ERA5 gain clearest at $h = 1$; **(b)** conformal coverage holds at the nominal 0.90 across horizons and predictor sets. Markers are means over three seeds.

Table S2: Occurrence feature-importance: L_1 -logistic stability-selection frequency (top 15 of $d = 49$ features), 95th percentile, $h = 1$. Lags are days; **exc30** is the 30-day exceedance count; **sin/cos_doy** are seasonality harmonics; “(mean)” denotes the daily mean of an ERA5 field over the synoptic box. Feature abbreviations follow the main paper (Section 3.3): IVT, integrated water-vapour transport; MSLP, mean sea-level pressure; NAO, North Atlantic Oscillation; the K-index is a convective-instability index.

London		Paris	
Feature	Freq.	Feature	Freq.
lag_1	1.00	lag_1	1.00
IVT (mean)	0.99	lag_6	1.00
cos_doy	0.99	K-index (mean)	1.00
sin_doy	0.99	lag_7	0.99
K-index (mean)	0.98	IVT east (mean)	0.98
500hPa geop. (mean)	0.98	MSLP grad. (E-W)	0.98
lag_5	0.95	lag_5	0.96
MSLP grad. (E-W)	0.93	cos_doy	0.96
exc30	0.91	NAO	0.93
lag_2	0.88	lag_14	0.93
lag_14	0.83	exc30	0.92
lag_8	0.82	lag_3	0.86
lag_12	0.82	MSLP grad. (N-S)	0.85
lag_9	0.82	lag_12	0.84
NAO	0.81	lag_2	0.84

Table S3: Occurrence Brier skill score (BSS) by forecast horizon, logistic +ERA5, 95th percentile. RES ($\times 10^3$) is the Brier resolution (informativeness); climatology has BSS = 0, RES = 0 at every horizon by construction.

Horizon h	London		Paris	
	BSS	RES	BSS	RES
1	+0.038	0.829	+0.117	5.820
3	−0.000	0.032	+0.048	2.435
7	−0.001	0.047	+0.032	1.746
14	+0.002	0.103	+0.017	0.708
28	+0.000	0.009	+0.015	0.685

Table S4: Coupling neither helps nor hurts: bake-off skill of the GP–EVT hurdle at the 95th percentile (+ERA5, $h = 1$, seed-averaged), with the coregionalised coupling on (as in the main text) and off (an independent hurdle). Brier skill score (BSS) and tail-CRPS skill (twCRPS) are relative to climatology; positive is better.

Model	London		Paris	
	BSS	twCRPS	BSS	twCRPS
GP–EVT hurdle (coupled)	+0.037	+0.032	+0.123	+0.094
GP–EVT hurdle (independent)	+0.037	+0.032	+0.127	+0.097